

Bridging the Digital Literacy Gap in Rural Agri-Logistics: A Trilingual, Context-Aware Intelligent Driver Assistance System for Sindh Supply Chains

Formal Research Proposal & System Architecture Document

Researcher: Lokesh Kumar (Student ID: 2k22-SE-42) | **Department:** Software Engineering, Sindh Agriculture University, Tandojam
Supervision & Guidance: Prof. Dr. Bhawani Shankar Chowdhry | **Live Deployment:** <https://agri-idas.tech>

1. Problem Statement & Motivation

In Lower Sindh (Tharparkar, Badin, Mirpurkhas), perishable agricultural produce such as tomatoes, chillies, and onions face over 30% post-harvest loss during transit to wholesale markets in Hyderabad and Karachi. Rural commercial drivers predominantly speak localized dialects (Sindhi and Dhatki) and cannot safely parse text-heavy English or formal Urdu navigation apps while driving. Furthermore, standard logistics systems rely on straight-line Euclidean or Haversine distance math, which drastically underestimates actual rural road distance along canal bunds and provincial highways, causing severe unbudgeted fuel deficits and cargo decay.

2. Core Research Objectives (RO1 - RO4)

- **RO1: True-Road GIS Modeling:** Replace naive Haversine approximations with Open Source Routing Machine (OSRM) graph traversal to capture actual road curves, bridge crossings, and monsoon detours across rural Sindh.
- **RO2: Low-Resource Dialect NLP:** Build an edge-compatible keyword and semantic intent parser specifically supporting the indigenous Dhatki dialect alongside Sindhi and Urdu.
- **RO3: Real-Time In-Cab Voice Guidance:** Deliver synthesized auditory advisories under 1.0 second turnaround so illiterate drivers receive prompt turn, speed, and hazard alerts without looking at screens.
- **RO4: Cold-Chain Telemetry Integration:** Link vehicle speed recommendations to real-time road friction ($\mu=0.38$ in rain) and refrigerated cargo temperature (19.2 C) to prevent tomato damage.

3. Technical System Architecture

Subsystem Layer	Technology / Framework	Operational Role in Rural Supply Chain
GIS Routing Engine	OSRM & OpenStreetMap (OSM)	1,313-point road-centerline navigation (Mithi -> Hyderabad corridor)
Trilingual NLP Engine	Phonetic Dialect Parser (Dhatki/Sindhi/Urdu)	Interprets vernacular driver distress calls and destination inquiries in < 0.01 ms
Auditory Voice Synthesis	Web Speech API / gTTS Speech Engine	Speaks natural voice advisories in Sindhi, Urdu, and Dhatki in ~747 ms
Context Safety Matrix	IoT Sensor Model (Friction μ , Temp, Diurnal)	Enforces 4-tier speed safety: Standard (72 km/h), Warning, Critical, Extreme (42 km/h)
Web Portal Interface	HTML5, CSS3, Leaflet.js, GitHub Pages	Live corporate fleet monitoring and driver cab display at https://agri-idas.tech

4. Empirical Findings: GIS Corridor Testing (Mithi to Hyderabad)

Corridor ID	Transit Corridor Name	Haversine (km)	True-Road (km)	Error (%)	Extra Fuel (L)	Unbudgeted Cost
R01	Mithi Farm A -> Mithi Market	1.49	1.99	25.00%	+0.06 L	Rs. 18.00
R02	Mithi Depot -> Naukot Junction	32.38	38.43	15.74%	+0.76 L	Rs. 213.26
R03	Naukot Belt -> Digri Market	56.24	75.04	25.05%	+2.35 L	Rs. 662.70
R04	Digri Tomato Belt -> Matli Hub	10.63	14.07	24.45%	+0.43 L	Rs. 121.26
R05	Matli Market -> Hyderabad Hub	62.91	70.01	10.14%	+0.89 L	Rs. 249.92
R06	Mithi -> Hyderabad Corridor (NH-8)	162.01	184.14	12.02%	+2.77 L	Rs. 779.88
R07	Diplo Pastoral -> Mithi Market	37.56	40.98	8.35%	+0.43 L	Rs. 120.39
R08	Tando Ghulam Ali -> Tando Jam SAU	49.03	58.21	15.77%	+1.15 L	Rs. 323.59
TOTAL / MEAN	Aggregate Regional Corridor Metrics	52.78	60.36	17.07%	+8.84 L	Rs. 2,488.81 PKR

5. Latency Benchmarks & Statistical Validation

- **Dhatki Intent Classification Speed:** Mean latency across 50 trials was **0.0034 ms**.
- **Audio Voice Alert Turnaround:** Mean end-to-end voice synthesis was **747.92 ms** (95th percentile: 766.12 ms), conforming strictly to ISO vehicular safety thresholds (< 1.0 second).
- **Paired t-Test Significance:** Distance discrepancy between Haversine and True-Road is statistically significant ($t = 4.892$, $p = 0.0017$, $p < 0.01$).
- **Survey Scale Reliability:** Pilot evaluation with 25 commercial truck drivers confirmed high internal consistency (Cronbach's $\alpha = 0.842$).

6. Supervisory Guidance & Verification

This research is being conducted as an official Final Year Software Engineering capstone and IEEE conference paper under the supervision of **Prof. Dr. Bhawani Shankar Chowdhry** at Sindh Agriculture University, Tandojam. The live interactive prototype, GIS simulation datasets, and complete codebase are accessible at <https://agri-idas.tech>.